

A Rayleigh Lower Bound for Normalized Inverse-GCD Matrices on Distinct Prime Pairs

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Disclaimer. Restricted Rayleigh bound only; not a full-spectrum floor. No Navier–Stokes claim.

Abstract

Let $\tilde{Q}_N(i, j) = 1/(\gcd(i, j)\sqrt{ij})$ for $1 \leq i, j \leq N$. For distinct primes p, q set $v = e_p - e_q$. We prove

$$R(v) = \frac{v^\top \tilde{Q}_N v}{\|v\|_2^2} = \frac{1}{2} \left(\frac{1}{p^2} + \frac{1}{q^2} \right) - \frac{1}{\sqrt{pq}} > -\frac{1}{2}.$$

We do not claim a full-spectrum floor $\lambda_{\min}(\tilde{Q}_N) > -1/2$ (false for moderate N), nor multi-representation bounds.

1 Operator

$$\tilde{Q}_N(p, p) = \frac{1}{p^2}, \quad \tilde{Q}_N(p, q) = \frac{1}{\sqrt{pq}} \quad (p \neq q \text{ primes}).$$

2 Theorem

Theorem 2.1. *For distinct primes p, q ,*

$$R(e_p - e_q) = \frac{1}{2} \left(\frac{1}{p^2} + \frac{1}{q^2} \right) - \frac{1}{\sqrt{pq}} > -\frac{1}{2}.$$

Proof.

$$R + \frac{1}{2} = \frac{1}{2} + \frac{1}{2p^2} + \frac{1}{2q^2} - \frac{1}{\sqrt{pq}} > \frac{1}{2} - \frac{1}{\sqrt{pq}}.$$

Since $pq \geq 2 \cdot 3 = 6$, one has $1/\sqrt{pq} \leq 1/\sqrt{6} < 1/2$, hence $R + 1/2 > 0$. □

Remark 2.2. *The Rayleigh quotient may be negative (e.g. $(p, q) = (2, 3)$ gives $R \approx -0.228$) but remains strictly above $-1/2$. Full-spectrum $\lambda_{\min}(\tilde{Q}_N)$ falls below $-1/2$ already for small N ; the present bound is restricted to single prime-pair difference vectors.*

3 Open

Uniform control for multi-representation vectors $v = \sum_a (e_{p_a} - e_{q_a})$ over unordered prime partitions of a fixed even integer remains open. Numeric checks through $N = 200$ stay above $-1/2$ on sampled multi-rep vectors; that does not constitute a proof.